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MSc Project Report

Fractal Modelling of Stock Market Time Series

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DECLARATION

This report is submitted to the Institute of Telecommunications (King's College London) as part of the course and is solely the original work of its author, except where clearly specified otherwise. It has not been previously submitted to this or another university.

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ABSTRACT

Stock markets have thwarted many years of efforts by mathematicians and statisticians to discover a recurring pattern hidden among their behaviour. The rationality was that if a pattern did exist then the traders, arbitrageurs(people who engage in the purchase and sale of securities within markets to make profits) would exploit this and quickly eliminate the chance of the pattern to continue existing or recurring. If the economic time-series data were assumed to be non-linearly dependent, rather than independent as the “Efficient Market Hypothesis” states, then much of the erratic behaviour can attributed to a deterministic response. This project aims at showing that the average trading volume of stocks that exhibit fractal properties have a functional relation with their fractal dimensions. We analyse this result by estimating their Hurst parameters by regression techniques for individual economic time-series of every stock in the NYSE that best fits the ‘fractal’ model. Then we find its correlation with the corresponding ‘average volume traded’. In the end a weak yet statistically significant correlation was found between the fractal dimensions and the average volume traded of the corresponding stock.

ACKNOWLEDGMENTS

I would like to utilize this opportunity to express my sincere gratitude to my supervisor, Dr. Enzo De Sena for his undivided attention towards the project and for his ceaseless motivation to do a better job. Dr. Enzo was instrumental in helping me find the right resources for my project and never let me stop till we achieved the relation we set out to establish. He was patient and supportive throughout the duration of my work, and offered immense technical support with MATLAB.

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[1]. INTRODUCTION:

"The supposition that the future resembles the past is not founded on arguments on any kind, but is entirely derived from habit, by which we are determined to expect for the future the same train of objects to which we have been accustomed" [1].

Complex systems exhibit complex characteristics such as sensitive dependence on small disturbances, long term memory, extreme variations and non-stationarity. Fractal phenomena are situations where a part of an object or phenomenon is exactly or statistically similar to another part or to the whole. Many time series, profiles and surfaces have been identified to exhibit fractal phenomena and are characterized by their fractal-dimension (a measure of their surface roughness) and by their Hurst parameter (a measure of long-term memory dependence). Both these phenomena have been modelled and elaborated by self-affine random functions, namely $1/f$ Gaussian noise and fractional Brownian motion (fBm) [2]. The assumption of statistical self-affinity implies a linear relationship between fractal dimension and the Hurst parameter, thereby linking these two phenomena [2]. Hence in our project we first set out to calculate the Hurst parameter of the economic time series in question to establish its fractal dimension.

It has long been known that the time series of stock-markets depicting returns have a self-similar structure, i.e. they look similar at different scales (years, days, minutes…). In other words they exhibit fractal properties. This feature can be used to gain insight into their

assumed chaotic behaviour and, to some extent, predict their short-term trend, as the search for predicting a trend or winning trade strategies has been the goal of every study in finance. The aim of the project is two-fold: first we evaluate how well the fractal model fits the time series of different types of stock prices and stock indices. (Here the word “model” shall not denote the mathematical expression of an economic relationship but will denote the statistical algorithm meant to satisfy to produce sample data streams that are hard to distinguish from actual records by eye or by algorithm [3].) This is validated by regression plots and the distribution self-similarity in Q function curves. Second we will identify correlations between the model parameters and the stock properties, such as trade volume categorized by market sectors.

Since fractals and fractal properties form the base of our research an introduction to fractals, self-similarity and fractal geometry is provided in Chapter 2. In this chapter we also discuss some methods to generate fractal images. Then we introduce the concepts of scaling and theory of finance in Chapter 3, we also talk about relation of fractals and stock markets and how fractional Brownian motion (fBm) helps in understanding the financial time-series. In Chapter 4 we introduce the methods and concepts required to carry the mathematical calculations required for this project and also discuss the design of MATLAB functions. Finally we analyse, discuss and validate the observations and results obtained in Chapter 5, providing suggestions for future work in this research.

[2]. FRACTALS AND FRACTAL GEOMETRY:

[2.1]. FRACTALS AND SELF-SIMILARITY:

Fractals are a relatively modern invention, since they have been recognized by mathematicians and scientists only over the last 40 years. Fractal are defined as a set that shows irregular but self-similar features on all or many scales. When an observed pattern of an image is similar to the shape of the image magnified a 100 times it makes it virtually impossible to determine which level or scale you are observing. Objects, shapes or a plots which follow this phenomenon are said to exhibit fractal properties. Another definition for fractals is as follows, it's a rugged or fractured geometric shape that can be broken down into subparts, each of which is a reduced copy of the whole.

The word fractal was first used by mathematician Benoit Mandelbrot; he based it on the Latin word, $fractus$ which means fractured or broken [4]. Fractals however are defined as a set, a mathematical set that usually exceeds the topological dimension, and the dimensional values may be fractional. Fractals paved to the concept of self-similarity, which states that "they are as same as when they were near, as they are from far" [4]. However on a broader scale the definition of fractals goes beyond the field of self-similarity. The oddity of fractals further intensifies as one starts to analyse it mathematically. To start with a fractal equation is not differentiable in any limit and also it can be perceived as a curve through space that despite being one-dimensional, resembles a surface. Fractals are not only limited to geometrical patterns, but are also definitive time dependent processes. In this thesis we discuss the concept

of fractals by using mono-dimensional fractals which was first introduced to the mathematical and financial world by Benoit Mandelbrot and the previous theories that failed and led to acceptance of fractals to define the economic time-series.

[2.1.1]. CHAOTIC SYSTEMS AND FRACTALS:

The Mandelbrot set is done with the creation of series of complex variable using the equation of the form $z_{n+1} = f(z_n)$. In the following example, also called as the Mandelbrot set the characteristic function is given by $f(z_n) = z_n^2 + z_o$ which yields a point on the complex plane for every initial point z_o on some division [4]. Fractal images can be generated on computer screens using various mathematical algorithms to generate them.

For this we introduce a concept called 'orbit visualizers' using strange attractors. An attractor is a set towards which a variable following the dynamical system (a system where the time dependence of points on an n-dimensional place relies on a fixed rule) evolves over time. The evolving variables are represented algebraically as an n-dimensional space vector [11]. Or in simple words the attracting subset of a phase space is a closed set to which many choices to an initial point in a system would have evolved to [11]. When the attractors evolve to or close to formal fractal structures then they are termed as 'strange attractors'. In 1986 Barry Martin developed the '$hopalong$' technique to demonstrate strange attractors of a chaotic system. For instance by using a series of two correlated equation comprising of complex variables generated on a plane we can generate the Julia set, where each coordinate generated is from a series represented as a point, or we '$hopalong$' from one generated point to another.

The following is a $signum$ function to demonstrate the $hopalong$ attractors:

$$x_{n+1} = y_n - sign(x_n)|bx_n - c|^{\frac{1}{2}}$$

$$y_{n+1} = a - x_n$$

This series of x, y coordinates is specified by an initial point x_o, y_o and three constants a, b and c. **Figure 1** is an example of Julia Set created from the above equations.

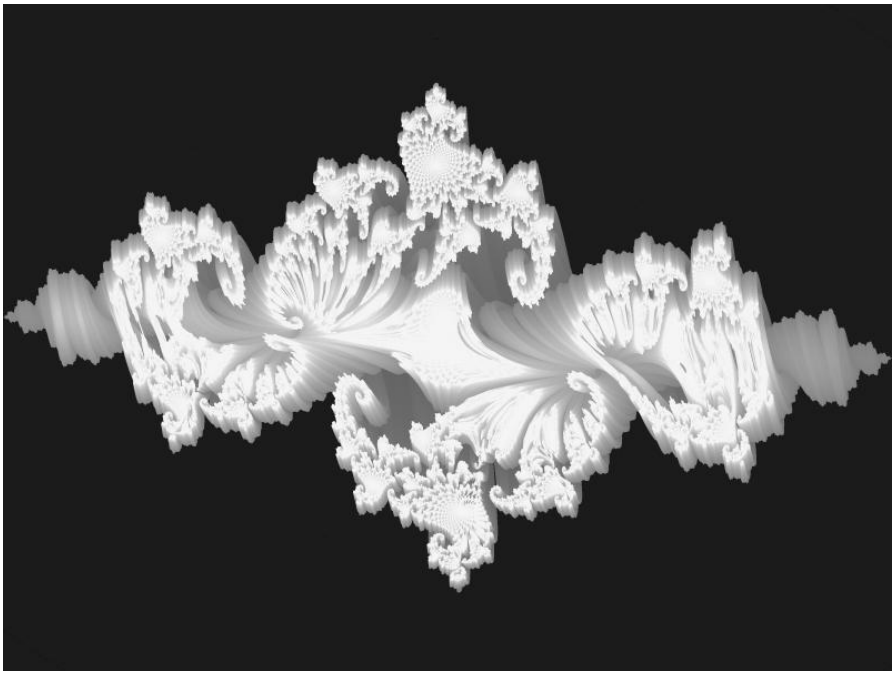

FIGURE 1. *An image of a Quaternion Julia set [10]*

[2.1.2]. NEWTON'S METHOD AND FRACTALS:

Newton Raphson method of finding solutions to polynomial equations results in some fractal images when applied to complex functions given in the form

$$f(z) = a_0 + a_1 z + a_2 z^2 + \cdots + a_m z^m = 0$$

The method generates a series where the n+1th approximation to the solution is given by

$$z_{n+1} = z_n - \frac{f(z_n)}{f'(z_n)}$$

It starts with an initial guess for the solution which may be any number and the Newton Raphson method gives you a close estimate, the subsequent estimates will be even closer to the solution. To create a two-dimensional image using this technique we start with a point which is somewhere between the solutions instead of starting close to the points [7].

For example consider a cubic polynomial $p(z) = z^3 - 1$, the newton function for this polynomial works out to be $f(z) = \frac{2z^3+1}{3z^2}$.

The transformation $\rho(z) = ze^{i2\pi/3}$, (where $e^{i2\pi/3}$ is a zero of the function $p(z)$) is a 120^0 rotation [8], resulting in the figure below. This transformation yields **figure 2.**

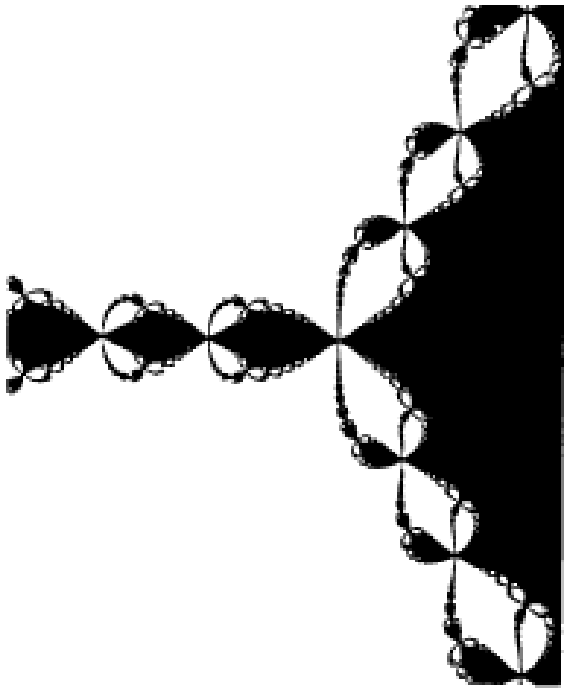

FIGURE 2. *Fractal image generated by Newton's method, adapted from [8].*

[2.2]. FRACTAL GEOMETRY:

Given a brief introduction about fractality and ways to generate fractal images we shall discuss the defining parameters of fractals or fractal geometry. Every geometric shape known to man or that observable in our daily life can be classified as with an integral dimension. Dimensions can be defined as the number of integral parameters required to define a curve or a surface or a shape with a definitive structure. For example when you consider a straight line, we term it as one-dimensional curve since we only require one parameter to uniquely define any point on the line. Similarly this applies in accordance to defining a curve, the circumference of a circle, or the perimeter of an object.

The mathematical definition of dimension defines how the scaling of an object varies with the linear dimensions. Consider a one dimensional line segment, if the linear parameter or the dimensional value defining the line segment is doubled then the length (characteristic size) of the line has to be doubled. In a two dimensional case, if the parameters defining a shape for example a rectangle is doubled, then the area is quadrupled [4]. In a three dimensional case for example a cube if the parameters defining the side is doubled then the volume increases by a factor of 8. This observation helps us to arrive at a relation between the dimension D, the linear scaling parameter S and the resulting increase in size V.

$$V = S^D$$

The equation explicitly defines what we already notice on observation [4]. Rearranging the above gives an expression, we can establish a relation for the dimension depending on how the size of an object changes as a function of linear scaling.

$$D = \frac{log(V)}{\log(S)}$$

In the examples described before the value of D are of integral values such as 1, 2, or 3, corresponding to the line, square and the cube. Every solid Euclidean object satisfies this relation and we can determine their dimension if we know their scaling parameters and their characteristic size. However, there many shapes or objects which do not agree to the integer based idea of dimensions or the relation defined above, i.e., there are objects which appear to be curves when observed superficially but the points on these curve cannot be uniquely described with integral number of parameters. There are shapes that lie on a plane but if they are linearly scaled by a factor S, the area does not increase by S^2 but by some non-integral value. These geometries are called fractals [4].

One of the fundamental fractal shapes is the **von Koch snowflake**. [9].

FIGURE 3. *The basic shape for the von Koch snowflake [8]*

To generate fractal patterns from such a basic shape one has to just perform simple iteration procedures of placing the von Koch snowflake in different orientations. Something similar is demonstrated in **figure 4.**

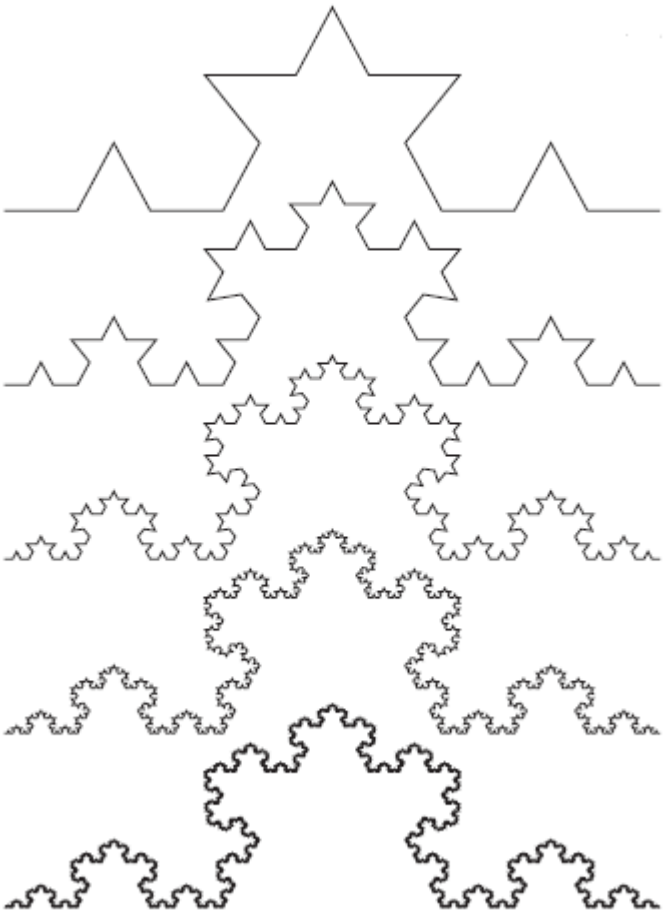

FIGURE 4. *When the line segments as shown in Fig – 7 is repeated for many iterations till it forms the shape of a snowflake, adapted from [8]*

This procedure demonstrates how a very simple generation rule with a fundamental shape can generate some unusual images with fractal properties. Unlike Euclidean shapes this object has detail at every level of scaling [4]. If one magnifies a geometrical shape to many folds such as the curve of a parabola it takes a different shape and reduces in dimension, namely a straight line. However if we were to magnify a fractal image unlike Euclidean shapes, further detail is revealed with every magnification, the detail is almost self-similar or rather it is exactly self-similar [9]. Note also that in **Figure 4** the shape generated in the end is not entirely a fractal but only a figurative demonstration of one. After every iteration the length of the shape increases by a factor >1 (4/3 to be exact) [9]. Considering the intuitive definition of one-dimensional shape when we apply it to the infinitive curves generated by the von Koch snowflake we notice that we cannot define a point on the curve with just one parameter as we don't know where it starts or ends thus giving it the name 'infinitive curve'. Although the algorithm for creating this "curve" is straightforward, there are no algebraic formulae that describes the points on the curve [9]. Some significant differences between fractal geometry and Euclidean geometry are listed below.

Fractal	*Euclidean*
Modern invention	*Traditional*
No specific size or scaling	*Based on characteristic size or scale*
Appropriate geometry in nature	*Suits description of man-made objects*
Described by a recursive algorithm	*Described by a usually simple formula*

TABLE – 1. *Differences between Fractal images and Euclidean shapes [9].*

[3]. fBm AND SCALING IN FINANCE:

[3.1]. INTRODUCTION:

Now that we have discussed fractals and their defining parameters, we introduce concepts of randomness, invariance, stationarity and scaling and how they have given way to the concept of fractals. We all know that randomness creates curious regularities or similar patterns but how to identify the interesting ones or the patterns which are of importance in a study especially in the field of economics. “To be random” is always perceived as something that lacks physical structure or property that would help an object to stand out among its kind. The standing question however is that whether this is a fair characterization of randomness or is the mathematical approach of chance meaningful enough to bring the strong degree or irregularity and variability encountered in financial charts. Mathematicians hold a broader outlook towards randomness than just fair coin tosses or other probability statements [3]. These mathematicians however deal with the observations that deviate around a median state that represents equilibrium. In the past random processes could only be investigated through formulas and theorems.

[3.2]. STOCK MARKETS AND DAY TRADING:

Now for an insight into the second element in this project or the element on which the fractal studies is conducted, **the stock markets**. A stock market is a public entity for trading of company stocks or shares and derivatives at an agreed price, in other words these are securities which are listed in a stock exchange. The **NYSE (New York Stock Exchange)** is an example of such market. Since at stock exchanges all the buying and selling happens in one place the traders can observe the fluctuating trends like the price of a stock in real time. The market traders include individual investors, institutional investors such as mutual funds, banks hedge funds, insurance companies and publically traded corporations trading in their own shares. The traders or investors can monitor how the stock price of a particular commodity behaves to news from the company, media reports and other influencing factors. Market pricing isn't entirely a mystery even with the fluctuating pricing, and spontaneous characteristics of the markets; it is influenced by the rise and fall of demand. Every day the trading begins with a fixed price on the value of all stocks called the **'Open Price'** and they vary for individual stocks. Throughout the trading day the prices vary or remain constant briefly depending on the volume traded and other factors listed above. The final price at which the stock is traded on the given day is defined as the **'Close price'** which is a measure to calculate correlation among various scale ranges and we calculate the Hurst parameter. The close price helps analyse the overall change in the value of the stock from when it started trading for that day and help the investor determine the trend followed by a particular stock or share.

Since trading in stocks is usually a high-stake affair, people who invest or trade are very careful and always seek methods to predict the future prices or extrapolate existing trends to predict the market behaviour.

Traders usually follow the “Portfolio theory” in selecting their stocks, assets or bonds to invest. The Portfolio theory states that the expected rate of return of a portfolio (a selection of options available for trading) is equal to the weighted sum of expected return of each individual option. It also states that the risk involved in trading can be reduced if the portfolio is diversified. The theory also suggests that that there exists an entirely risk-free asset for every portfolio to counterbalance the risk.

Another theory which most of the traders and stock market analysts believe in is the “Efficient Market Hypothesis”. According to which a market with random price movements indicate a well-functioning market rather than an irrational one [10]. The efficient market hypothesis maintains that markets prices reflect all the trading information on a stock, including its randomness. In an efficient market, the price changes must be unpredictable even if the information and expectations of all market participating stocks are incorporated [11].

Other than the “portfolio theory” and the “efficient market hypothesis” there are various other models and theories looming around the stock markets. Some even help decide the pricing options and to invest in the right stocks for the right amount. For example, the “Black-Scholes” equation and the “Binomial Options Pricing Model” both are a set of equations which takes price of the derivative, the risk free interest rate and the volatility factors as inputs to calculate the estimate option pricing and rate of return.

[3.3]. RELATION BETWEEN FRACTALS WITH FINANCIAL MARKETS:

Every investor, professional currency or stock trader knows and has experienced inexplicable changes in daily price quotations. Fortunes are made as quickly as they are lost in stock trading on account of these volatile bursts. For example the stock price of Alactel a French telecommunications equipment manufacturer dropped to about 40% in a day and further deteriorated the next few days. On the fourth day, the trend reversed and it shot up by 10%. Several other examples like the housing bubble in the year 2005-06 lead to the stock market crash in 2007 and the credit defaults by major banks led to the stock market crash in 2008-09, these incidents were totally unforeseen. The conventional financial models, like the portfolio theory and the effective market hypothesis speculate that these events should never had happened. These risk reducing formulae suggest that price changes are statistically independent of each other or that they follow the Martingale property, which is successive price changes are mutually uncorrelated [12], as a result there is no scope for prediction of future market movements. Second presumption is that these price changes follow a bell curve distribution, such that the width of the bell curve depicts the probability of such large deviation from the mean occurring which is negligible for such drastic events mentioned before. It is evident that the modern portfolio theory and other risk-reducing theories pose a risk to the investors who strongly believe in them. Over the years revisionists and mathematicians have tried to correct the objectionable assumptions of the portfolio theory. A proper quantitative study of financial markets began in the 1960's earlier to which the study of finance relied on the "1900" model of Bachelier, which pointedly denied discontinuity (sudden change in trend of trading prices) and concentration (prices changes do not happen evenly over time, the rise and fall of prices are concentrated in short turbulent periods) [13]. To tackle

discontinuity and concentration the concept of self-affine was associated with the financial markets. The notion is more important, and also most visual hence making it self-explanatory, in the context of financial charts. Two renormalized (rescaled) charts are never identical of-course the belief was that they do not differ in kind. The scholarly term to "look alike" is "to remain statistically invariant by dilation or reduction." Viewing the renormalized pieces of chart (as shown in **Figure** 5 in the following page) as statistical samples from the same underlying process, several non-Brownian motion of scaling were identified one after the other [9].

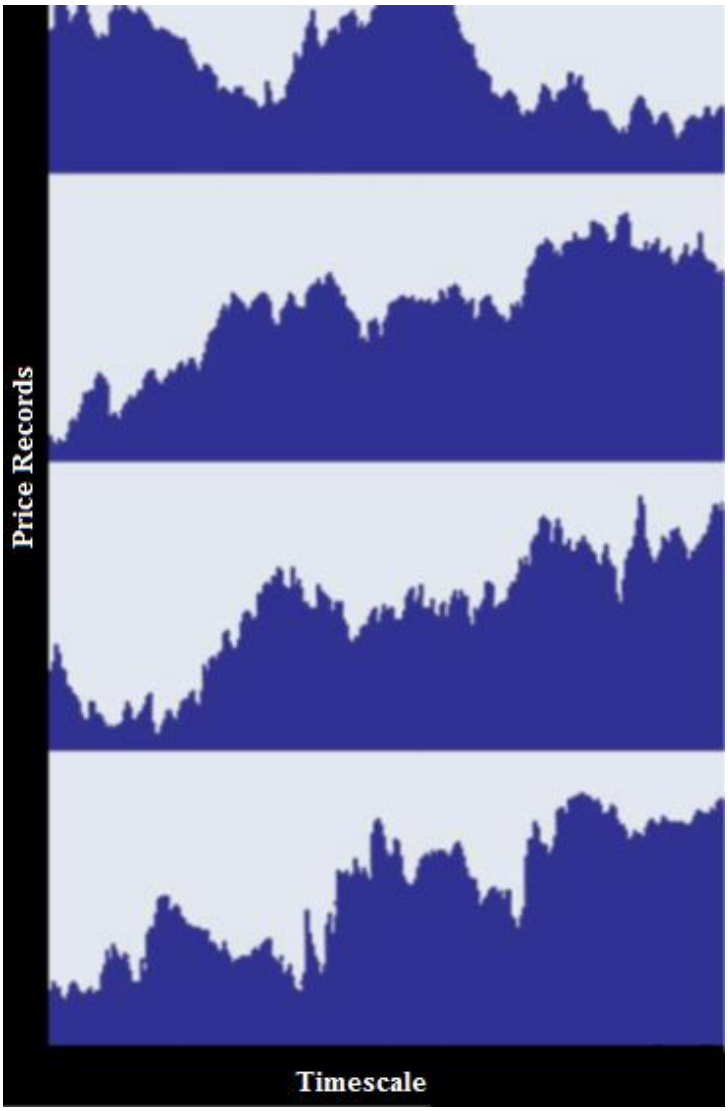

FIGURE 5. *A collection of diagrams, illustrating the behaviour of at least one actual financial price at various timescales. Adapted from [14].*

Note:

It is easily acceptable that randomness can create its peculiar regularities or patterns, but difficult to acknowledge or study these regularities that could be of interest in the field of physics or economics. The problem is aggravated in economics because each entity is known individually can be hampered with easily. Most seek financial knowledge for the purpose of finding a trend and benefitting from it thus modifying what they benefit from.

[3.4]. BACHELIER'S THEORY OF SPECULATION:

Consider a time-series of prices denoted by $Y(t)$ and the $Z(t)$ be its logarithmic relative

$$Z(t,T) = log_e Y(t,T) - log_e Y(t)$$

The basic price model variation assumes that successive increments $Z(t,T)$ are

(i) Random

(ii) Statistically independent

(iii) Identically distributed and

(iv) Gaussian with zero mean

This process is called the stationary Gaussian random walk or the Brownian motion. Although this model continues to be important its assumptions must not be converted into doctrines. Bachalier himself made no mention of empirical evidence in favour of Brownian motion. He noted that his model contradicts the evidence in two ways [13]. Firstly, the sample variance of $Z(t)$ varies in time, he attributed this to variability of population variance. Second, Bachalier noted that no mixture of Gaussian distributions could account for sizes of the very large price

changes, he treated them as “outliners”. Thus he revealed the major weakness of the Gaussian random walk model. However new advances in the theory of speculation continues to be best expressed as improvements upon the existing Brownian motion model.

[3.4.1]. THE 1900 MODEL:

Drawing analogy between “The random walk” and the price variations:

The 1900 model assumes that the price variation is random and that each price change is statistically independent of its previous instances. The continuous counterpart of random walk was advanced to Brownian motion.

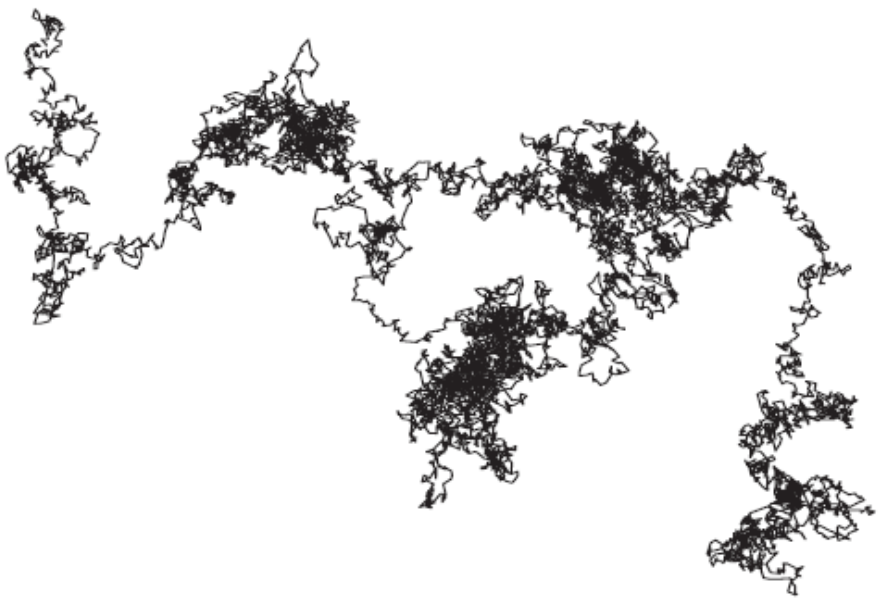

FIGURE 6. *Planar Brownian motion, adapted from [15]*

[3.4.2]. INVARIANCE PROPERTIES OF BROWNIAN MOTION:

Statistical stationarity of price variations: Equal intervals of a straight line can be perfectly superimposed on each other which is not possible for the plot of a random process. Brownian motion however defies the rule of a random process and exhibits properties of superimposition over equal time increments in a statistical sense.

Scaling of price: It states that the non-overlapping increments of variable durations in Weiner Brownian motion, can be superimposed when suitably rescaled [3].

The analogy of Brownian motion with price changes provides us with a few more properties of the market:

(i) Independence of price increments: Statements are similar to Martingale's theorem. Knowing the values of past instances can provide no knowledge of the future.

(ii) Continuity of price variation: Brownian motion samples display continuous curves even though they do not have any derivatives anywhere. This property almost surely holds well in the market scenario.

(iii) Rough evenness of price changes

(iv) Absence of clustering in the time of large changes & absence of cyclic behaviour

Another assumption on the Brownian motion $B(t)$ is the Martingale process. It states that: whether the prices of the past is known fully, partially or not at all, the price change over all future timespans have $E[B(t)] = 0$ [3]. However, this statement allows properties other than expectation $E(t)$ to depend on the past. However this assumption raises some serious difficulties. Firstly a 'positive' martingale converges always; i.e. it eventually settles down and

ceases to show randomness. Similarly, a martingale that continues to show randomness eventually becomes 'negative' martingale which implies a random walk becomes a negative martingale and we know that the logarithmic price scale is a Brownian motion and thus a martingale and since price range cannot be negative it ceases to be a martingale thus questioning the 'efficient market' hypothesis. Another issue arising is that while it ideally implements the efficient market, it is impossible to implement arbitraging even under ideal conditions.

Inadequacies in Brownian motion as a model of explanation:

Significant mathematical research grew around the Brownian motion model since it was much more manageable than any other alternative [3]. It also paved the way for the financial mathematics, which draws heavy reference from assumptions that prices follow martingales and that $B(t)$ applies for every price. However this was contradicted by Bachelier himself also saying that the data were non-Gaussian and statistically dependent. Various discrepancies were listed as research continued.

[3.4.3]. INVARIANCE PRINCIPLES: STATIONARITY AND SCALING:

The fractal model approach to finance and economics relies on two prominent features. First feature being the importance of invariances and possibility of identifying scaling and stationarity as invariances [3]. Second prominent feature is the recognition that probability theory is more versatile than the general consensus and its willingness to work with several distinct state of randomness [3]. In the field of mathematics such principles are key to a myriad of consequences drawn from one simple idea but they have not been imbibed by

economics as of yet. Hence, Mandelbrot suggests to drop Brownian motion as a model but to preserve the stationarity and scaling invariance principles which was elaborated.

[3.5]. FRACTIONAL BROWNIAN MOTION (fBm):

One of the most successful models for explaining random fractals in nature has been the fractional Brownian motion. It is an extension of the concept of Brownian motion that has helped define many phenomenon in physics and mathematics. There are two well established yet contradictory definitions of fractional Brownian motions, one prevailing in the probabilistic literature, the other in the econometric literature [16].

Each is associated with a different definition of non-stationary fractional time series, arising in functional limit theorems based on such series. It is the inherent chaotic nature of the economical time series of the returns that has allowed us to choose fBm to explain the phenomenon of self-similarity in the time series. We will see that combining the concepts of fractals with Brownian motion will lead to some intriguing results.

“fractional Brownian motions" (fBm’s), are defined by a family of Gaussian random functions as follows: $B(t)$ being ordinary Brownian motion, and H a parameter satisfying $0 < H < 1$, fBm of exponent H is a moving average of $dB(t)$, [16] in which past increments of $B(t)$ are weighted by the factor $(t - s)$. We believe fBm does provide useful models for a host of natural time series as previously mentioned.

An example of fBm curves with corresponding Hurst parameters are shown in **Figure 7** to demonstrate how the Hurst parameter affects the roughness of a curve.

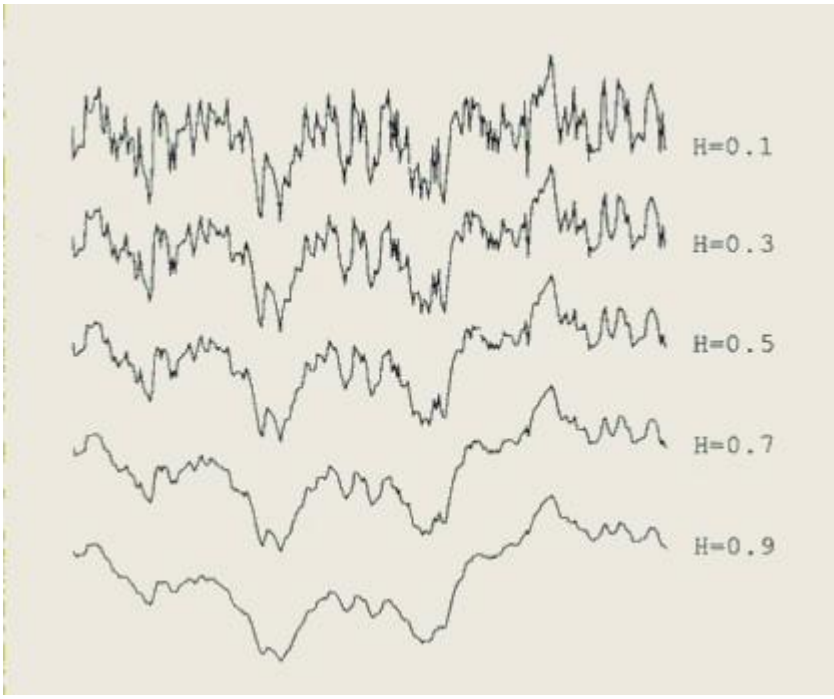

FIGURE 7. *Fractional Brownian motion curves with corresponding Hurst parameters, adapted from [2]*

The basic feature of fBm is that they can have infinite range of interdependence between their increments. Hence, the study of random functions has been exclusively dedicated to the sequences of independent random variables, Markov processes, and to other random functions having the property that sufficiently distributed samples of these functions in terms of distance of the ranges are independent, or nearly so [16]. Empirical studies of random chance phenomena often suggest a strong interdependence between distant samples which is quite contrasting [16]. It is known that economic time series "typically" exhibit cycles of all magnitude, the slowest cycles having periods of duration comparable to the total sample size.

[3.5.1]. DEFINITION OF FRACTIONAL BROWNIAN MOTION:

$\forall\, H \in (0,1)$ a Gaussian process $B^H = \{B_t^H \quad t \geq 0$ is a fractional Brownian motion if the mean expected value is 0 i.e. E(B_t^H) = 0 and covariance is given by

$$\mathrm{E}[B_t^H, B_s^H] = {}^{1}\!/_{2}\,[|t|^{2H} + |s|^{2H} - |t-s|^{2H}] \quad \forall\, t, s \in R$$

The standard Brownian motion eq. can be recovered by substituting $H = \frac{1}{2}$

Let H be such that $0 < H < 1$, and let b_o be an arbitrary real number. $B_H(t,\omega)$ or the reduced fractional Brownian motion, with coefficient H and starting value b_o at time 0. For t>0 $B_H(t,\omega)$ is defined by

$$B_H(0,\omega) = b_0,$$

$$B_H(t,\omega) - B_H(0,\omega) = \frac{1}{\gamma\left(H+\frac{1}{2}\right)}\{\int_{-\infty}^{0}[\,(t-s)^{H-\frac{1}{2}} - (-s)^{H-\frac{1}{2}}\,]dB(s,\omega) + \int_0^t (t-s)^{H-\frac{1}{2}}\,dB(s,\omega)\}$$

(and similarly for t<0). Point-wise integration is carried out using integration by parts [16].

[3.5.2]. SELF-SIMILARITY PROPERTIES:

$$\{X(t,\omega)\} \triangleq \{Y(t,\omega)\}$$

The above notation means that two random functions $X(t,\omega)$ and $Y(t,\omega)$ have the same finite joint distribution curves.

The increments of a random function $\{X(t,\omega);\ -\infty < t < \infty\}$ will be said to be self-similar $(s-s)$ with parameter H $(H \geq 0)$ if for any $h > 0$ and t_0

$$\{X(t_0+\tau,\omega) - X(t_0,\omega)\} \triangleq \{h^{-H}[X(t_0+h\tau,\omega) - X(t_0,\omega)]\}$$

Thus implicating that increment of $fBm, B_H(t,\omega)$ are stationary and self-similar with parameter H [16].

The correlation between two increments of $B_H(t,\omega)$. Let $T, T_1\ and\ T_2$ be fixed and non-negative. Now we compute the increments of this series following the time intervals $T/_2$ to T_1 and $-T/_2$ to T_2.

We arrive at the following correlation:

$$\frac{1}{2}\left(\frac{(T+T_1+T_2)^{2H} + T^{2H} - (T+T_1)^{2H} - (T+T_2)^{2H}}{{T_1}^H {T_2}^H}\right)$$

By replacing $S_1 = {T_1}/_T$ and $S_2 = {T_2}/_T$ so as to reduce the correlation as only a function of S_1 and S_2 [16]

For all S_1 and S_2, this correlation is positive for $\frac{1}{2} < H < 1$ and negative if $0 < H < \frac{1}{2}$

This gives rise to few properties of fBm

(i) If $0 < H < 1/2$ the series is said to display Short Range Dependence

(ii) If $H = 1/2$ then the time series has independent increments or the series is a std. Brownian motion

(iii) If $\frac{1}{2} < H < 1$ then the time-series displays Long Range Dependence. If a fBm exhibits Long range then it is safe to say that the series has memory.

[3.5.3]. ESTIMATING FRACTAL DIMENSION USING THE HURST PARAMETER:

From the previous section it can be said that '$hurst$' parameter is a dimensionless estimator of the self-similarity of a time-series. On the other hand the fractal dimension D, of a surface is the measure of roughness of the surface with $D \in [n, n+1)$ for a surface in n-dimensional space with higher values indicating rougher surfaces [17]. In principle the $hurst$ parameter and fractal dimension D are independent of each other, since fractal dimension is a superficial or surface property whereas long-term memory dependence is a global characteristic. However these two parameters are closely related in many scientific literature. The $hurst$ exponent is related to the fractal dimension D as $D = 2 - H$ ($D = n + 1 - H$) for an n-dimensional surface) [18].

In relation with the project:

As we have made a mention of the different mathematical proposals that lead to the acceptance of fractals as a plausible concept to explain randomness in financial markets, an extensive mathematical basis exists for fractal studies already. In finance this concept is not a baseless generalization but a theoretical reformulation of the market movements, especially the movements of stock or currency, which all appear to look similar when enlarged or reduced so as to accommodate all scale ranges. An observer cannot tell the scale ranges of a graph (**Figure 8)** simply by looking at it.

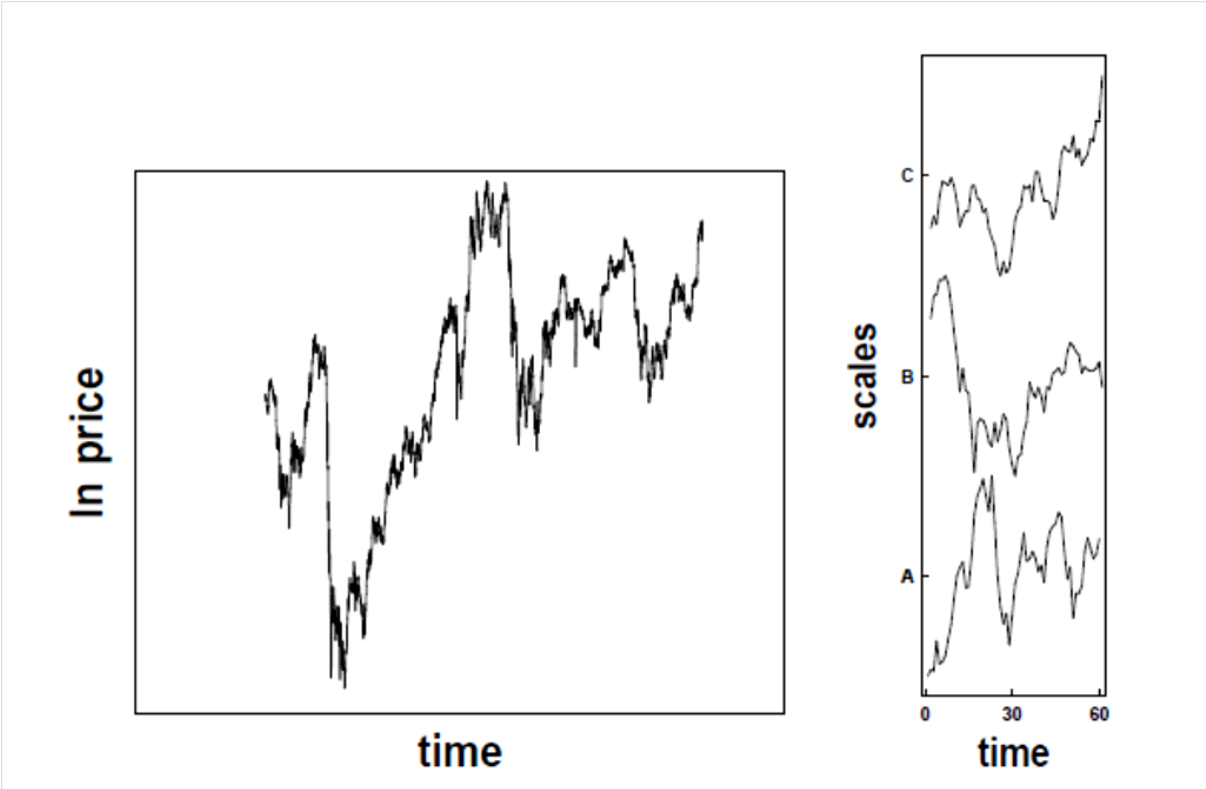

FIGURE 8. *left) natural log of daily prices right) a 60 day daily price record, a 60 week weekly price record and 60 month monthly price record, not depicted in the order dictated, adapted from [19].*

This indicates the fractal nature of the financial charts. The beauty of fractal geometry is that it can devise a model general enough to reproduce that are in accordance with the portfolio

theory and also include the erroneous trading incidents. Sums of Random variables and their properties play an important role in the modelling of stock markets. To proceed we assume that the logarithm of tomorrow's price $\ln z(t+1)$ that of today's price $\ln z(t)$ plus a random price increment [20]. This log price series $\ln z(t)$ fits well in a self-affine fractal geometry framework. Such geometries are described by the H parameter which attributes the surface fluctuations. We use the fact that the difference of the inter-quote prices is independent of physical time-interval between successive quotes, therefore to estimate the hurst exponent we calculate the mean of the absolute values of the moving average of price records over varying interval τ given by $< |y(t+\tau) - h(t)| > \sim \tau^H$ [19] where τ is a measure of lag in terms of quotes. The log of this equation yields a correlogram which forms the base for our calculations and to estimate the hurst parameter for finding correlations. However the hurst exponent gives limited information about the underlying distribution of differences in the log prices for different time-scales. The distribution and its scaling properties as a function of the log of τ or the lag can provide a much more quantitative characterization.

It is important to note that increments $\boldsymbol{Z}_K = \boldsymbol{Y}(t+k) - \boldsymbol{Y}(t)$ of fBm of type H are stationary and have a Gaussian distribution with variance depending on time scale k where $Y(t)$ is the logarithmic price value [20].

$$\boldsymbol{Y}(t+k) - \boldsymbol{Y}(t) \text{ i.d. } \left(\frac{k}{k'}\right)^H \big(Y(t+k') - Y(t)\big), \ldots\ldots\ldots \text{ [19]}$$

Where i.d. denotes identically distributed. Here $k^{-H}(Y(t+k) - Y(t))$ converges to a distribution. However, the convergence is not to a stable distribution of index $\frac{1}{H}$, but to the Gaussian. A quantitative test for the self-similarity in a time-series is to estimate the mean of the size of the jumps across time-lags k, and to look at the scaling behaviour. For the economic time series the lag is measured in units of business days. The time series is then $\{Y(i)\}_{t=1}^{T}$, where $Y(t)$ is the log of the close price on business day t.

The estimates of corresponding Hurst scaling exponent H

$$< |\boldsymbol{Y}(t+k) - \boldsymbol{Y}(t)| > \sim k^H \qquad \text{(eq. 1)}$$

In order to analyse that the distribution of absolute of the log of returns are scale invariant, we apply following modification to reflect the scale invariance.

$$\frac{Y(t+k) - Y(t)}{k} \;\; \boldsymbol{i.d.} \;\; (\frac{k}{k'})^{\boldsymbol{H-1}} \frac{Y(t+k') - Y(t)}{k'}$$

The $p.d.f.$ of these rate of returns over time periods of size k is denoted by $P_k(\rho)d\rho$ .For densities that satisfy

$$(H-1)\ln k + \ln P_k(\rho) = \ln Q(k^{1-H}\rho),$$

implies that the process is self-similar if plots of $P_k(\rho)$ collapse onto function Q, when plotting [19].

$$(H-1)\ln k + \ln P_k \;\; \text{Versus} \;\; k^{1-H}\rho$$

In this case the distribution of the returns on all scales k is fully determined by the "basic distribution" of Q, and the scaling exponent H.

While performing the above calculations of correlations and correspondingly finding the slope against the log of the interval, we follow a heuristic approach which is we limit the max size of the increments k to a specific value (75% of the length of the time-series). This is to eliminate an anomaly arising due to the fact that statistical mean or the expected value (comprising of all the observers in the dependent variable also referred to as the population mean) $E[|\boldsymbol{Y}(t+k) - \boldsymbol{Y}(t)|]$ isn't the equivalent of sample mean $< |\boldsymbol{Y}(t+k) - \boldsymbol{Y}(t)| >$ which comprises of only a set size of observers. Simply because to calculate a sample mean we need a considerable

number of points and if we don't limit the size of the increments as the k value increases the number of points to calculate the sample mean $< |\boldsymbol{Y}(t+k) - \boldsymbol{Y}(t)| >$ decreases.

[4]. METHODS, RESEARCH & DESIGN:

[4.1]. REGRESSION TECHNIQUE - METHOD OF LEAST SQUARES:

Now to estimate the $hurst$ parameter as mentioned earlier by using the log of correlation points and the log of size of time lags k we use linear regression techniques.

There are various linear regression techniques, for this project we are using the **method of least squares** to find the best fit model and to calculate the slope along with the intercept. The technique uses simple calculus and line algebra for finding the best fit straight line of the form $y = ax + b$ given that for $n \in \{1,2, \dots N\}$, (x_n, y_n) are the observed points. In addition to this we also calculate the chi-squared error which specifies the deviation of the fitness of the curve from the data points [21]. The basic idea of any least square fit is to find the model which minimizes the sum of the vertical distances squared between all data point and the least square model. The method generalizes to finding the best fit of the form

$$y = a_1 f_1(x) + \cdots c_k f_k(x);$$

To obtain the estimates of the coefficients, the least-squares method minimizes the sum of square of the residuals [21]. The residual for the ith coordinate r_i is the difference between the observed response value y_i and the fitted response value $\hat{y}_i$, and is identified as the error associated with the data.

Given a set of data $\{(x_1, y_1), \dots, (x_n, y_n)\}$ the error associated to the best fit line $y = ax + b$ is given by

$$E(a, b) = \sum_{n=1}^{N} (y_n - (ax_n + b))^2$$

Which is in-fact just N times the variance of the data set $\{y_1 - (ax_1 + b), \dots, y_n - (ax_n + b)\}$

The objective is to find the values of a and b that minimizes the error.

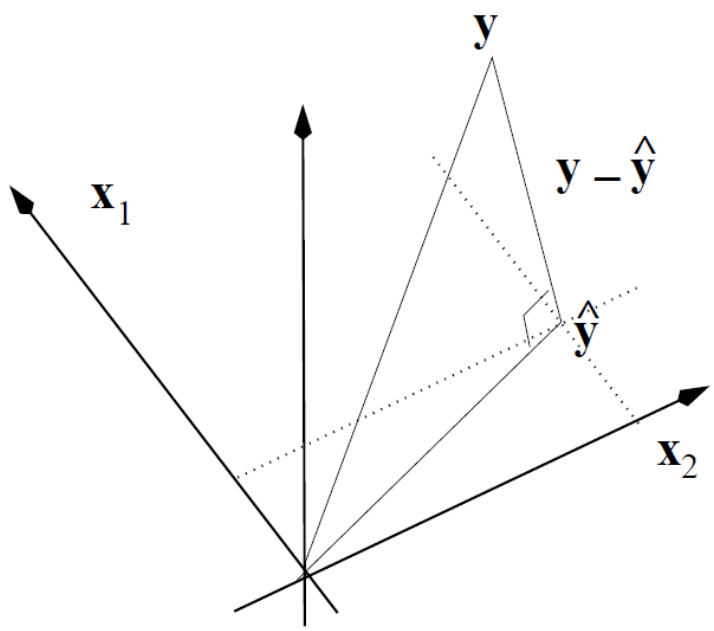

FIGURE 9. *Residuals demonstrating least squares method, adapted from [22]*

$$r_i = y_i - \hat{y}_i$$

$$residual = data - fit$$

A *linear* model is defined as an equation that is linear in the coefficients. To illustrate the linear least-squares fitting method, consider *n* data points that can be modelled by a first-degree polynomial.

$$y = ax + b$$

To solve this equation for the unknown coefficients a *and* b, we write E as a system of n simultaneous linear equations with two unknowns. If n is greater than the number of unknowns, then the system of equations is termed as *over-determined*.

$$E = \sum_{i}^{n} (y_i - (ax_i + b))^2$$

Because the least-squares fitting process minimizes the summed square of the residuals, the coefficients are determined by differentiating E with respect to each parameter, and setting the result equal to zero [21].

$$\frac{\partial E}{\partial a} = -2 \sum_{n=1}^{N} x_n \big(y_n - (ax_n + b)\big) = 0$$

$$\frac{\partial E}{\partial b} = -2 \sum_{n=1}^{N} \big(y_n - (ax_n + b)\big) = 0$$

$$\sum x_n \big(y_n - (ax_n + b)\big) = 0$$

$$\sum \big(y_n - (ax_n + b)\big) = 0$$

where the summations run from i = 1 to n. The $normal\ equations$ are defined as

$$a \sum x_n^2 + b \sum x_n = \sum x_n\, y_n$$

$$a \sum x_n + Nb = \sum y_n$$

Solving for a

$$a = \frac{N\sum x_n y_n - \sum x_n \sum y_n}{N\sum x_n^2 - (\sum x_n)^2}$$

Solving for b using the a value

$$b = \frac{1}{N}\left(\sum y_n - a\sum x_n\right)$$

As you can see, estimating the coefficients a and b requires only a few simple calculations, where a is the '$hurst$' parameter or the slope we are trying to find and b is the y intercept.

Next we estimate the **goodness of fit or the $R^2 statistic$,** it is a measure of how well the least square fit of a line follows the trend in the data [23].

$$R^2 = \frac{\sum(\hat{y} - \bar{y})^2}{\sum(y - \bar{y})^2} = 1 - \frac{\|r\|_2^2}{\sum(y - \bar{y})^2}$$

When $R^2 \approx 1$ the goodness of fit for a particular is accurate, however

When $R^2 \approx 0$ the goodness of fit is insignificant in this case [23].

Before explaining the next phase of the project or as to how the results obtained are processed it is mandatory that the reader is exposed to a concept termed as the null hypothesis.

[4.2]. NULL HYPOTHESIS:

The null hypothesis significance testing (NHST) is a dominant approach to statistical inference in any research. This method has been often challenged, occasionally defended and has been persistently used in many scientific and statistical research studies [2]. NHST is the primary means by which the data for an experiment is regarded as a conclusion about the objective that the experiment was originally designed to address. The first basic significance testing can be traced back to 1710 [26], however modern significance testing has developed since 1900. Karl Pearson, formulated the first modern significance testing (the chi-square goodness of fit test) [25]. In statistical reference the null hypothesis implies that there is no correlation between two dependant variables which is the general or default position. Thus rejecting or contradicting the null hypothesis implies that there is a correlation among the variables in our case the measured average of the difference between the time series of an interval k. There are two ways to deal with the null hypothesis, one with 'statistical significance' and other with 'hypothesis testing' approach. In 'statistical testing', a null hypothesis is usually rejected or disproved. In the 'hypothesis testing' the null hypothesis is usually contradicted with an alternative hypothesis. Conjoining both these approaches in a single problem is both incorrect and incoherent.

The null hypothesis is never proved or established, but is possibly disproved in the course of the experiment [24].

The null hypothesis significance testing works as follows: first we measure the set of data and its correlation with another assumingly dependent variable and calculate the probability of how unlikely the particular result is assuming the null hypothesis is true.

Another question arises as to how do we measure the probability of the unlikeliness of the event occurring? Here we introduce another concept which we use in this project: Pearson Correlation.

[4.3]. PEARSON CORRELATION & p-VALUE:

The null hypothesis works by assessing whether the calculate deviation measured by the test statistic is usually larger than a pre-defined value so that the probability of the occurrence of this assumption is very small under the null hypothesis (usually less than 5%). If the particular data set doesn't reject the null hypothesis, then only a weak conclusion can be made, i.e. the observed data set provides no strong claim to reject the null hypothesis, mostly because the data is insufficient to reject the hypothesis.

The Pearson product-moment correlation coefficient is a measure of linear correlation (dependence) of two random variables X and Y. It takes values from [-1, 1]; -1 indicating a strong negative correlation and +1 indicating a strong positive correlation. However, to calculate the probability of an assumption or to the test the hypothesis we use the p-value generated for the Pearson correlation test for testing the hypothesis of no correlation against an alternative hypothesis that there is a non-zero correlation. To compute the p-value we require a null hypothesis to reject or accept, a test static and data. The p-value for the Person-Product correlation moment is calculated as follows

$$s = \frac{r * \sqrt{n-2}}{\sqrt{1-r^2}}$$

where r is the correlation coefficient & n is the number of data pairs.

Traditionally we can reject the null hypothesis if the p-value is less than or equal to a specified significance level often 0.05 [25].

In our project we define the null hypothesis to reject as: "the correlation between the $hurst$ parameter or the fractal dimension values and the average of the volume traded for corresponding stocks is 0". Thus upon calculating the Pearson correlation coefficient and the subsequent p-value we will establish the degree of linear correlation (dependence) and the probability of rejecting the assumed hypothesis.

[4.4]. MATERIALS AND MATLAB FUNCTIONS:

To proceed with the project and to calculate the Hurst parameter of the stock values for various stock indices is to be downloaded. The data should include, trading dates, open price, close price, volume and other parameters needed for finding correlations. The required data is extracted from Yahoo! Finance, a website that holds a database of stock quotes, exchange rates, corporate press releases and financial reports. These historical data of a particular stock are available in daily, weekly, monthly and dividends only quotes. The price quotes are updated with just 15 minutes of delay. There are a total of 2643 stocks trading in the NYSE and Yahoo! Finance has reliable data for only 2338 stocks. Next step lies in extracting the data for these stocks using MATLAB, to isolate the Close price for every stock and to perform correlation studies to estimate the Hurst parameters for these stock indices using the regression analysis techniques.

Yahoo! finance provides a CSV file for the historical stock data of any desired Stock index, provided it has the information in its database which can downloaded onto any system. The URL for downloading the stock-data csv of the following format

http://ichart.finance.yahoo.com/table.csv?s=**MSFT**&a=**08**&b=**7**&c=**1984**&d=**01**&e=**31**&f=**2013**&g=d&ignore=.csv

On comparing the parameters selected in the site and the designated URL to download the CSV file we can deduce what each parameters stand for in the URL, or in other words we can break the URL so as to implement it in MATLAB.

It pertains to the following format

http://ichart.finance.yahoo.com/table.csv?s="**TICKR_SYMBOL**"&a="**START_MONTH**"&b="**START_DAY**"&c="**START_YEAR**"&d="**END_MONTH**"&e="**END_DAY**"&f="**END_YEAR**"&g=d&ignore=.csv

If the data requested is beyond the range of the historical prices in the Yahoo! Finance database then the url simply returns all the data available within the specified data range.

Next task is to construct the desired url to download the CSV stock-data files for various stock indices. This is done via MATLAB's 'urlread' command.

Then the individual data elements of the CSV file are stored into an array of structures. This is demonstrated by the following code snippet.

```
stock_data(y).Ticker = tickSym;                    % obtain ticker symbol
stock_data(y).Date = date;                         % save date data
stock_data(y).Open = str2double(open);             % extract open price
```

```
stock_data(y).High = str2double(h);                    % extract high price
stock_data(y).Low = str2double(l);                     % extract low price
stock_data(y).Close = str2double(close);               % extract close price
stock_data(y).Vol = str2double(vol);                   % extract volume
stock_data(y).AdjClose = str2double(a_close);          % extract adjusted close
                                                         price
```

This is included in a function that performs the task of accepting the start date, end date and tick_sym and feeding it into the 'urlread' command to retrieve the data and this function is called recursively for all the stock indices stored as 'char' array.

Struct_array(idx) = stock_price_data (**'START_DATE', 'END_DATE', 'TICK_SYM'**);

This line is included in 'try', 'catch' blocks so that if the stock-data for a particular stock isn't available, it simply ignores that index and continues with the next 'TICK_SYM' without terminating the iteration process.

Now that we have the array of structures for stock data, we need to extract the 'close price' to calculate the correlation and to calculate the slope of their regression curve (which we perform using the method of least squares as we discussed earlier) for various timescales or intervals. Once the close price is extracted the values in the variable are flipped so as to store it conventionally in the order of oldest price of stock to the current trading price.

Self-similarity in an economical time series follows the scaling behaviour of absolute size of jumps across lags of size k [19]. The first quantitative test for self-similarity to estimate the mean of the size of the jumps across the time lags of size k and to look at the scaling behaviour of this quantity as a function of k [19]. Log function is chosen out of customary reasons for economical time-series/price-records.

Let $Y(t)$ be the log of the closing price of any desired stock, it is measured in terms of number of quotes and not the time interval of successive quotes.

As depicted in equation (1) to estimate the $hurst$ parameter or the scaling exponent we need to find the slope of $\log k$ vs. $\log[< |Y(t+k) - Y(t)|>]$ will give the value of H or the hurst parameter [19].

Because the correlations play a very crucial role in shaping the basic distribution it seems more remarkable that the price records analysed here have large degree of distributional self-similarity [19]. This means that the market may actively and perhaps unknowingly self organizes itself so as to achieve distributional self-similarity.

Having said that now we shall get back to calculating the Hurst parameters first by calculating the correlations over various time lag sizes k.

```
correlogram(k) = mean(abs(Y((k+1):end)-Y(1:(end-k))));
```

using the above code segment we get the correlation points of the log price over various time intervals and is stored in an array. Here we limit the value of k to 75% of available scale range because of the discrepancies arising over lack of sufficient points to calculate the sample mean as we discussed in the section "**population mean vs. sample mean**".

The correlation coefficient and the 'rsquare' values are stored in an array and so is the average volume of the selected stocks for which the number of available data points are greater than 1000 for better computation of correlation. As discussed earlier the fractal dimension D of a particular stock is simply $(2 - h)$ value which are also stored in array for the multitude of stocks. Further we perform a histogram of the available h parameters or the slopes of the

correlations for each stock prices. To further arrive at a conclusive result we narrow our selection of stocks to those best fitting the regression model, i.e. we take only those stocks for which the rsquare values are > 0.9. Now that we have the correlation coefficient (slope), fractional dimensions and the average volumes of the restricted stocks we proceed to calculate the existence of correlation among them. For this we use the Pearson-product correlation coefficient and p-value.

[5]. RESULTS:

[5.1] OBSERVATIONS:

Stock data for ticker symbol 'AA' or Alcoa Inc.					
Date	Open Price	Highest Intraday Price	Lowest Intraday Price	Close Price	Volume Traded
'2013-08-01'	8	8.08	7.93	7.95	18071900
'2013-07-31'	8.02	8.08	7.94	7.95	14872800
'2013-07-30'	8.04	8.06	7.98	8.03	11533200
'2013-07-29'	8.06	8.06	8	8.04	8394100
'2013-07-26'	8	8.08	7.96	8.07	8731000
'2013-07-25'	8.04	8.11	7.93	8.03	12347000
'2013-07-24'	8.14	8.14	8.01	8.05	11937500
'2013-07-23'	8.15	8.21	8.1	8.14	13853800
'2013-07-22'	8.1	8.14	8.07	8.12	8124700
'2013-07-19'	8.23	8.23	8.04	8.07	23767400

TABLE – 2. *10 observations out of the 12985 observations for the stock 'AA' or Alcoa Inc.*

Table 2. shows the first 10 observations of the 12985 observations extracted from the Yahoo! Finance database for the stock 'AA' or Alcoa Inc. The downloaded data for the stock includes 12913 samples of 'Open Price', 'Close Price', 'Highest Intraday Price', 'Lowest Intraday Price' and 'Volume Traded' starting from 'Jan 1962' to 'August 2013' (time when observations were last updated). As mentioned earlier we have similar stock data available for 2379 stocks traded in the NYSE, starting from the earliest date available in the database until their last trading day.

Then the log of the close price is isolated into another variable for further calculations.

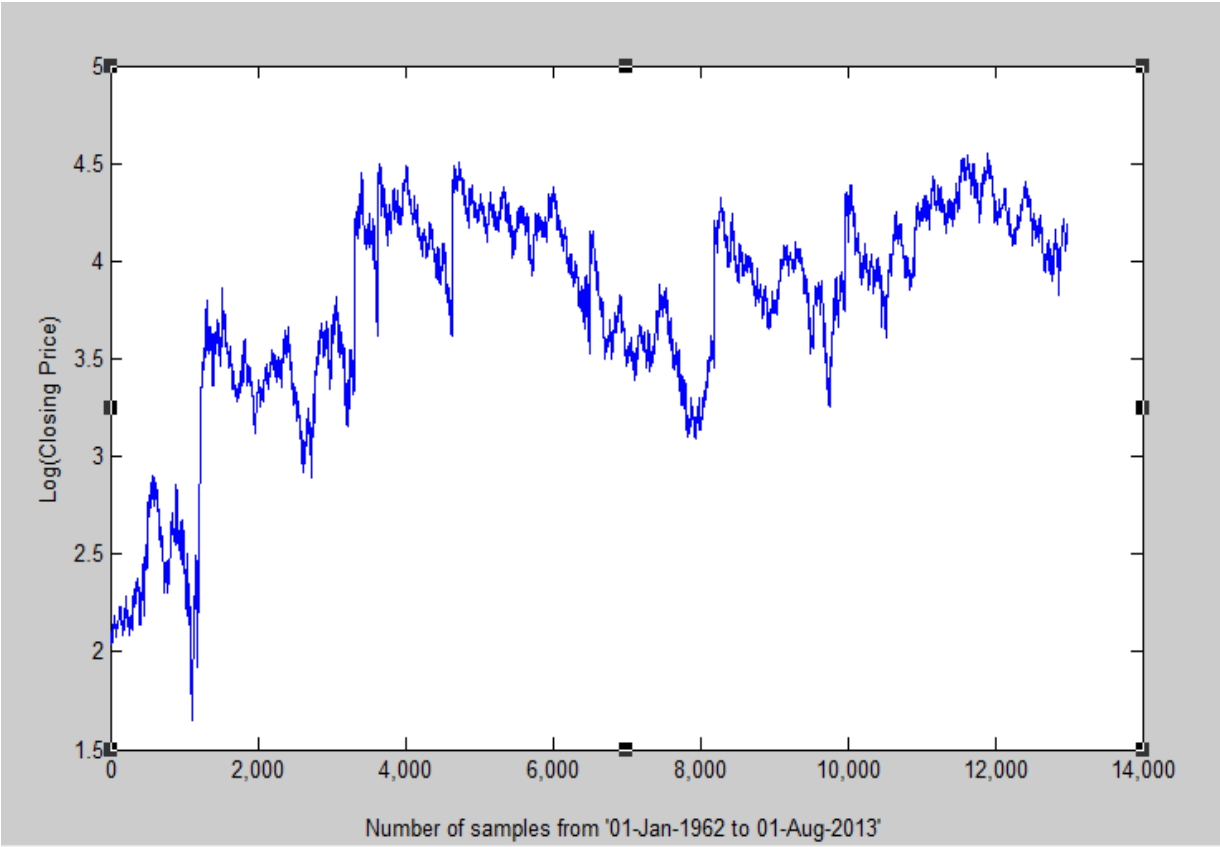

FIGURE 10. *A plot showing the log of Close price for stock 'AA'*

Figure 10 is an example to display the shape of the log of close price so that one may observe the recurring fractal patterns or self-similarity across the graph for various time intervals. In other words the time-series can be split into various intervals to notice that different segments superimpose with each other.

Next step is to calculate the correlogram for the stock prices based on **Equation 1** and to plot the log of correlogram with the log of number of samples (75% of available data) based on the heuristic approach.

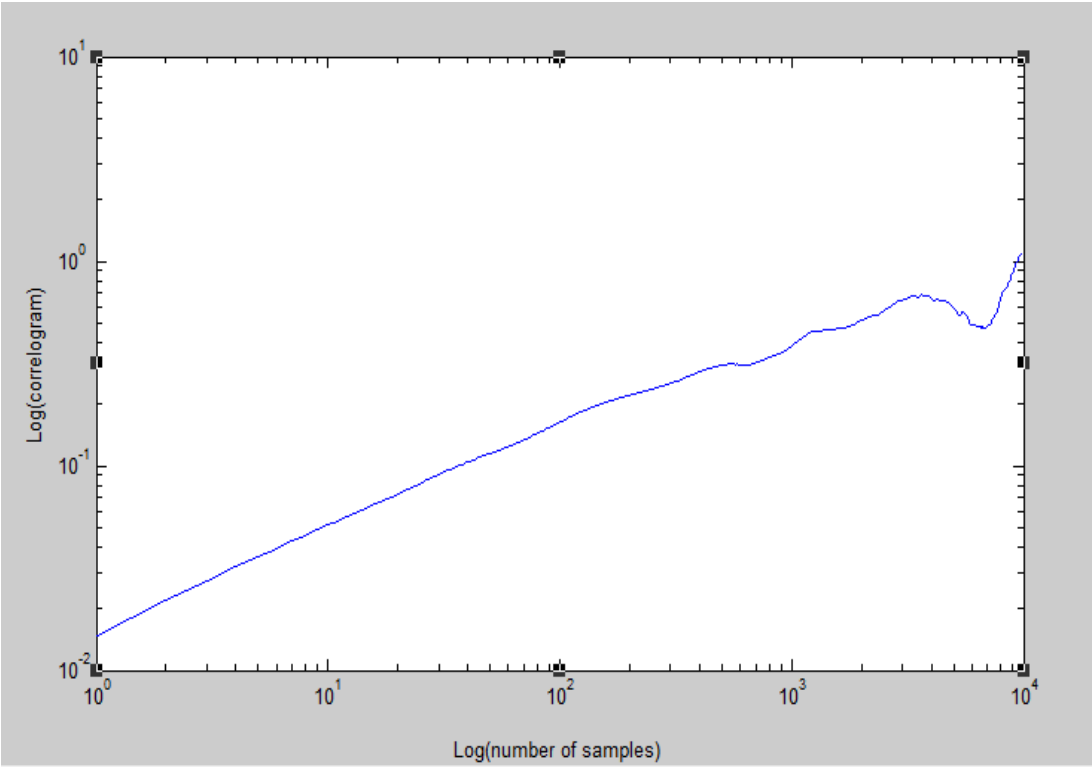

FIGURE 11. *A log plot of correlogram vs. number of samples.*

Figure 11 shows the log plot generated for the stock 'AA'

We then proceed to perform the regression analysis to calculate the slope of the correlogram (hurst parameter) using regression techniques. In MATLAB, the default regression technique used is the method of least squares.

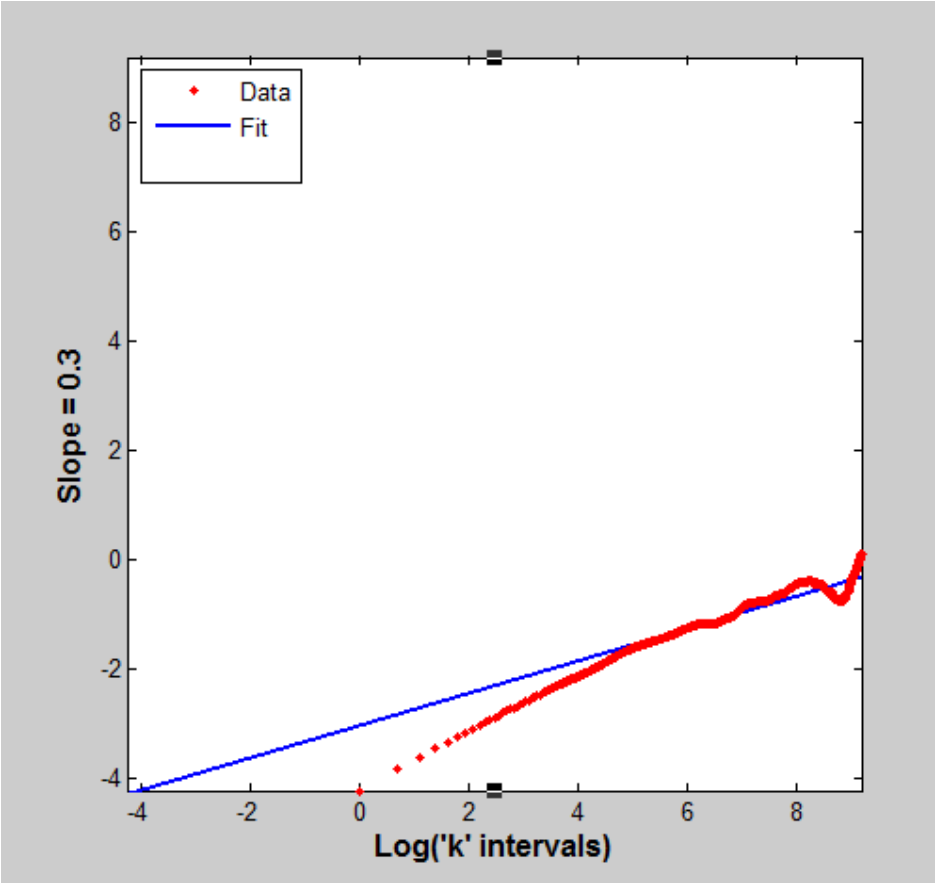

FIGURE 12. *Regression plot of log (correlogram) vs. log (sample size) for stock 'AA'*

In **Figure 12** as denoted in the legend, the red dots represent the data points to which we try to map a best fitting line, to calculate the slope. The blue line is the best fit curve for the data points and the slope of this line is the approximate slope of the curve formed by the data points or in other words, the hurst parameter. The goodness of fit or the rsquare value would be around 60%-70% for this case.

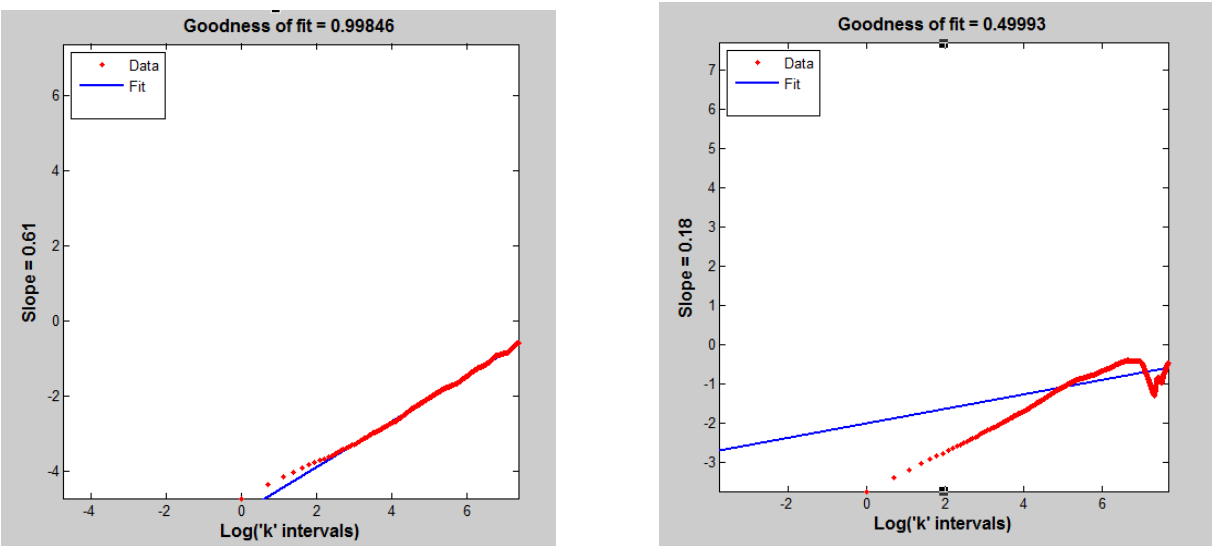

FIGURE 13. *left) Example of a best fit curve (stock 'MSP') right) Example of a bad fitting curve (stock 'ABG')*

As described in **Figure 13** the left hand side plot is an example of the best fit curve, in fact that plot has the highest rsquare value of the goodness of fit (99.85%). On the contrary, the right hand side plot demonstrates a curve that has the worst fit among the stocks (49.99%).

The slope of the correlogram for various stocks (hurst parameter) is stored in an array and the corresponding fractal dimensions are calculated and stored in a separate array.

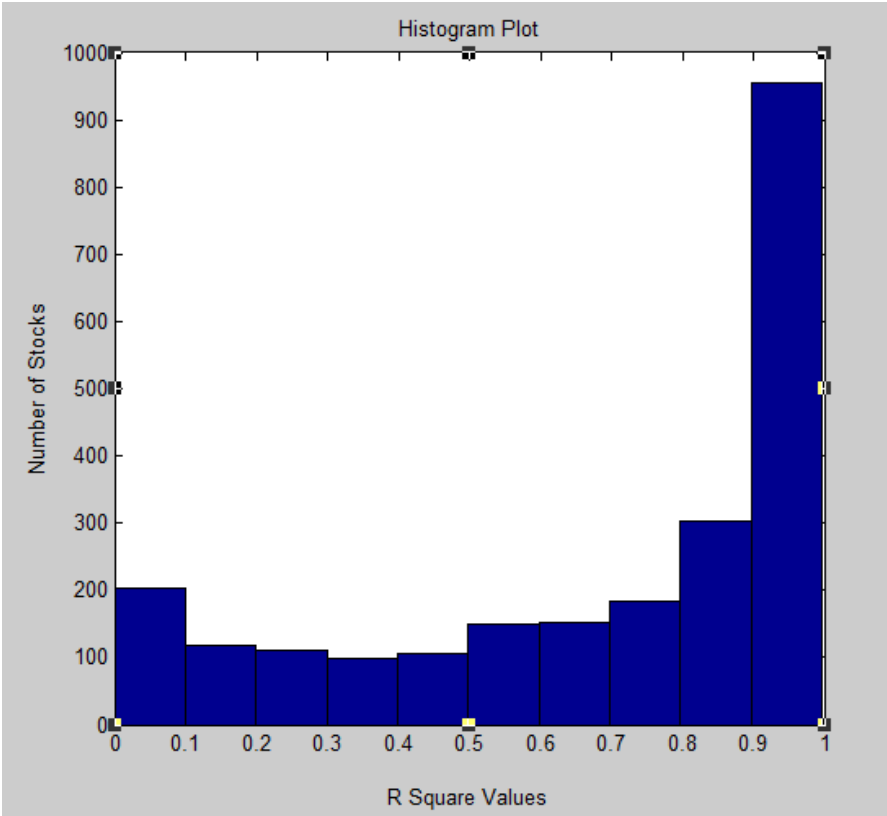

FIGURE 14. *A histogram plot of the fractal 'rsquare' values*

It is evident from **Figure 14** that we have sufficient number of stocks that are well fitting ($rsquare > 0.9$) the fractal model. So the Pearson-product correlation coefficient between the fractal dimensions and average volumes are evaluated for these well-fitting stocks and the p-value is calculated to test the strength of the correlation.

The correlation value was found to be 0.0931 and the p-value was estimated as 0.043. The importance of these results are elaborated in the 'Discussions' section.

To further strengthen the claim of the correlation between average volume traded and the **fractal dimensions** we group the stocks according to their average volume traded and we find the average fractal dimensions for each group. **Then we do a regression plot to find if there is a linear relation among them.**

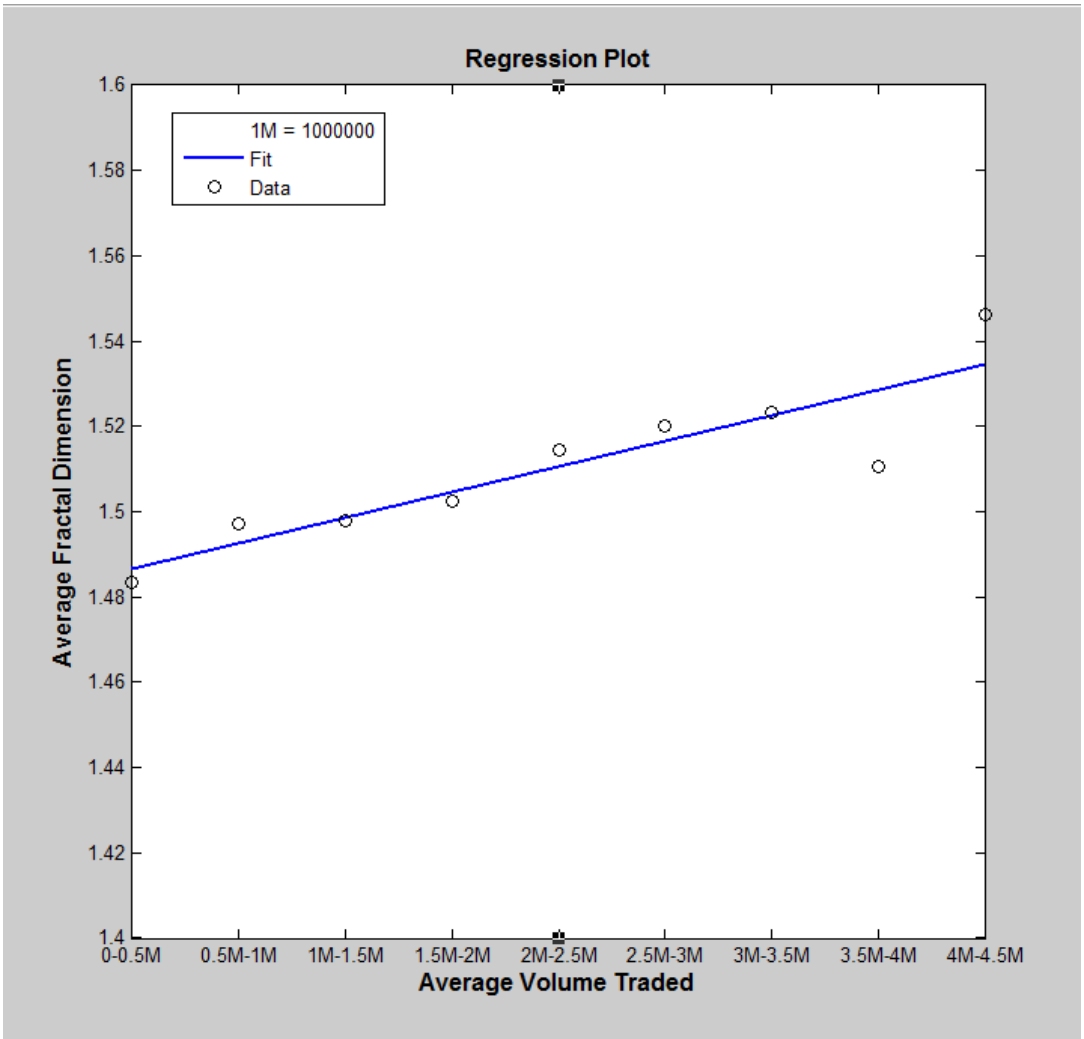

FIGURE 15. *Regression plot showing a linear relation among the fractal dimension and the average volume traded.*

From **Figure 15** it is evident that the **fractal dimension increase as the volume traded increases** thus establishing **that highly traded stocks exhibit high variability** thus helping us attain the objective of this project.

Finally a scatter plot between the fractal dimensions and the average volumes for each stock is drawn.

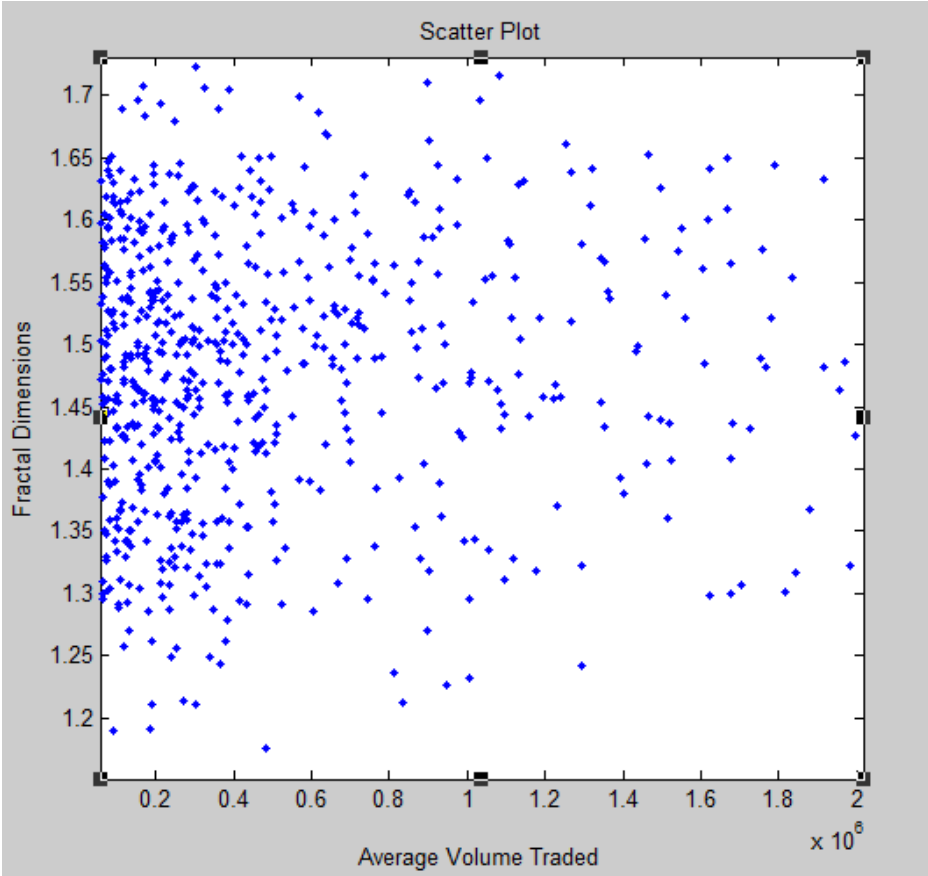

FIGURE 16. *Scatter plot of fractal dimensions vs. average volumes of selective stocks.*

The lack of any linear relation among the points in the scatter plot can be attributed to the very low value of the Pearson coefficient we observed in our results.

[5.2]. DISCUSSIONS:

We set out to extract the stock data from a particular stock market (we have chosen the **NYSE** for this project**)** and to estimate the hurst parameters for the log of economic time-series for individual stocks over various intervals 'k' through regression methods. Due to sheer volume of stock data available and the various timescale ranges they're available in, it is difficult to generalize and to obtain the result we set out to achieve, hence we set some constraints like limiting the size of 'k' intervals for a time-series while calculating the 'correlogram' and also restricting the 'raquare' or the goodness of fit to values > 0.9.

Then we try to establish the correlation between the fractal dimensions and the corresponding average volume traded for these selective stocks. The **correlation coefficient** of fractal dimensions with the average volume traded for every stock results as **0.0931**, and the **p-value** was obtained as **0.043**, this establishes that the correlation between the two parameters is **existent but very meek**. We arrive at this conclusion based on the p-value because our null hypothesis was defined as "correlation between the parameters is 0" and since the p-value is < 0.05 we can reject this hypothesis thus concluding that the correlation is not zero but existent. The very low value of the Pearson correlation can be explained over the fact that Pearson-product correlation only searches for linear correlation, it may so be that the average volume traded and the corresponding fractal dimension of each stock may have weak linear correlation but a stronger relation in other functional forms (for example parabolic relation). However from the present linear correlation values obtained we can conclude that stocks trading in higher volume exhibit higher variability. In other words stocks with higher trading volume fit the fractal model better.

Accuracy of the project is dependent entirely on the stock data on Yahoo! Finance which quite is reliable and used by many traders for managing portfolios. The 15 minute delay in availability of trading prices is of no issue to the project thus not affecting the calculations involved.

To discuss further on the findings observed on the course of the project, it's best to begin with the impact of the financial crisis on the time series and correlation studies. To investigate this the time-series was divided into pre-financial crisis and post-financial crisis intervals and a difference in behaviour was observed, suggesting that the financial crisis is a statistically significant anomaly and further research can be carried on in this regard and observe of inclusion or exclusion of various occurrences of such disturbances. Also the research can be further extended to other stock markets and the stocks can be split according to their sectors to try and observe distinctive fractal patterns or correlation between the trading parameters.

[5.3]. CONCLUSIONS:

The project aimed at calculating the fractal dimensions of various timeseries plots and also at finding a relation between the calculated fractal dimensions and any trading parameters of a stock (in this project we chose the average volume traded). Upon careful selection of stocks that best fit the fractal model (or that has the higher rsquare values which determines the goodness of fit) and grouping those according to the range of volumes traded and by plotting them against their corresponding average fractal dimension we observe a closely linear relation thus achieving the aim of the project.

REFERENCES:

[1]. D. Hume: *A Treatise of Human Nature.* (1978) Glasgow, Scotland: William Collins (Original work published 1739).

[2]. J. Krueger: "Null Hypothesis Significance Testing: On the survival of a flawed method" (2001), Brown University.

[3]. B. B. Mandelbrot: "Fractals and Scaling in Finance: Discontinuity, Concentration, Risk" (1997), Springer-Verlag, New York
ISBN 0-387-98363-5, pp. 26-79

[4]. B. B. Mandelbrot: *The Fractal Geometry of Nature* (1982) W. H. Freeman, San Francisco

[5]. Auslander, J., Bhatia, N.P., Seibert, P.: "Attractors in dynamical systems." (1964), Bol. Soc. Mat. Mex.9, pp. 55–66.

[6]. Von D. Ruelle: *Chaotic Evolution and Strange Attractors.* (1989.) Cambridge University Press, Cambridge (UK), Vol. VIII

[7]. A. Burton, "Newton's Method and Fractals" (2009), *Journal Article*

[8]. K. Falconer: *Fractal Geometry: Mathematical Foundations and Applications*, 2nd Ed. (c) 2003, John Wiley & Sons Ltd.

[9]. H. -O. Peitgen, Dietmar Saupe, M. F. Barnsley, R. L. Devaney, B. B. Mandelbrot, R. F. Voss, Y. Fisher, and M. McGuire "The Science of Fractal Images". Springer-Verlag, New York, 1988. ISBN 0-387-96608-0.

[10]. Z. Bodie, A. Kane, & A. J. Marcus: *Essentials of Investments: Global Edition*, pp.228-229

[11]. A. W. Lo: Efficient Market Hypothesis, *The New Palgrave: A Dictionary of Economics*, 2nd Ed. 2007. NY: Palgrave McMillan

[12]. B. B. Mandelbrot: "The Variation of Certain Speculative Prices" (1963), The Journal of Business, 36, 394-419.

[13]. L.Bachelier: *Theory of Speculation* thesis (1900)

[14]. B. B. Mandelbrot: "Scaling in Financial Prices: I. Tails and Independence" (2001), *Quantitative Finance,* Vol.1, pp.113-123.

[15]. P. Morters & Y. Peres: *Brownian Motion* , 2010, Cambridge University Press, NY, pp.2

[16]. B. B. Mandelbrot and J.W. van Ness: "Fractional Brownian motions, fractional noises and applications", *SIAM Review* (1968) pp.422-437

[17]. E.Bayratkar, H. V. Poor, K. R. Sircar: "Estimating the Fractal Dimension of the S&P 500", (2002).

[18]. T. Gneiting & M. Schlather: "Stochastic models which separate fractal dimension and Hurst effect" (2001), *NCRSE-TRS,* Vol. 69

[19]. Carl J. G. Evertsz: "Fractal Geometry of Financial Time Series", *Fractals* (1995) Vol.3, pp.609-616

[20]. Carl J. G. Evertsz and K. Berkner: "Large Deviation and self-similarity analysis of graphs: DAX stock prices", *Chaos, Solutions & Fractals* (1995) Vol. 6, pp.121-130

[21]. S. J. Miller, "Method Of Least Squares" *Journal Article,* ID 5246752

[22]. H. Abdi: "Least Squares" (2003), *Encyclopedia of Social Sciences and Research Methods.* Thousand Oaks (CA): Sage.

[23]. G. W. Recktenwald: Supplement to *Numerical Methods with* MATLAB: *Implementations and Applications*, © 2000–2007, Prentice-Hall, Upper Saddle River, NJ.

[24]. C. Pesse: " Long Range Dependence of Financial Time Series Data" (2009), *International Journal for Human and Social Science*

[25]. T. R. Levine, R. Weber, C. Hullett, H. S. Park, & L. L. M. Lindsey: "A Critical Assessment of Null Hypothesis Significance Testing in Quantitative Communication Research", Human Communication Research ISSN 0360-3989, doi:10.1111/j.1468-2958.2008.00317.x

[26]. G. Loftus: "Null hypothesis", (2010) *Encyclopedia of Research Design*, SAGE Publications, Inc., Thousand Oaks, CA, pp. 939-943.

[27]. Carl J. G. Evertsz: Self-similarity of high frequency USD-DEM exchange rates, Proc. Of first Int. Conf. on High Frequency Data in Finance, Zurich 1995, Vol.3.

[28]. M. F. Guiducci and M.I. Loffredo: "Time Series Analysis: Mandelbrot Theory at work in Economics" (1997), *Transactions on Engineering Sciences* Vol.15, ISSN 1743-3533

[29]. T. Gneiting & M. Schlather: "Stochastic models which separate fractal dimension and Hurst effect" (2001), *NCRSE-TRS*, Vol. 69.

[30]. R. A. Thisted: "What is p-value?" (1998), *Dept. of Statistics and Health Studies, University of Chicago*

[31]. G. A. Edgar: *Measure, Topology, and Fractal Geometry* (2008), 2nd Ed. Springer

[32]. G. Gigerenzer, & D. J. Murray (1987). Cognition as Intuitive Statistics. Hillsdale,NJ: Lawrence Erlbaum Associates

[33]. T. R. Levine, R. Weber, C. Hullett, H. S. Park, & L. L. M. Lindsey: "A Critical Assessment of Null Hypothesis Significance Testing in Quantitative Communication Research", Human Communication Research ISSN 0360-3989, doi:10.1111/j.1468-2958.2008.00317.x

[34]. D. Ruelle: Chaotic Evolution and Strange Attractors (1989), Cambridge U.P.

[35]. E. Peters (1994): Fractal Market Analysis, John Wiley and Sons

[36]. P.H. Cootner: The random character of stock market prices (1964) The M.I.T. press, Cambridge, Massachusetts

[37]. G. R. Richards: "A Fractal Forecasting Model for Financial Time Series" (2004), Journal of Forecasting, J. Forecast. 23, pp. 587-692.

APPENDIX

MATLAB CODES:

Primary Code:

```
T = char (NYSE_symbols); %To store the Stock Symbols as 'char' array
l = 1;


for k = 1:length (NYSE_symbols)
        tickSym = T(k,1:end); %Extracting the stock symbols one at a time


        try
            stocks(l) = stock_price_data('01011950','01082013',tickSym); %To
store stock data in array of structures
            l = l+1;
        catch ME
            continue %To ignore faulty 'ticker symbol’
        end


end


stock_correlation_H_volume; %Invoking functions to estimate the correlation
```

To extract the stock data from Yahoo! Finance

```
function stocks = stock_price_data(dateBeg, dateEnd, tickSym)

stock_data = struct([]);        % initialize data structure

start_day = dateBeg(1:2);                           % beginning day
start_month = sprintf('%02d',str2double(dateBeg(3:4))-1);
% beginning month
start_year = dateBeg(5:8);                          % start year

end_day = dateEnd(1:2);                             %end day
end_month = sprintf('%02d',str2double(dateEnd(3:4))-1);
% ending month
end_year = dateEnd(5:8);                            % end year

% frequency of downloading stock data
f = 'd';                                        % default is daily

x = waitbar(0, 'Please Wait...');                   % create waitbar
y = 1;                                          % increment stock index position
```

```
% run iteration for each ticker and retrieve stock data
 % update waitbar to display current ticker
 waitbar((1-1)/length(tickSym),x,sprint('%s %s %s%0.2f%s', ...
    'Extracting stock data for',tickSym,'(',(1-1)*100/length(tickSym),'%)'))

[t, stats] =
urlread(strcat('http://ichart.finance.yahoo.com/table.csv?s='...tickSym,'&a=',st
art_month,'&b=',start_day,'&c=',start_year,'&d=',end_month,'&e=',end_day,'&
f=',...end_year,'&g=',f,'&ignore=.csv'));

 if (stats)
    % segregate data
    [date, open, h, l, close, vol, a_close] =
strread(temp(43:end),'%s%s%s%s%s%s%s',',');

    stock_data(idx).Ticker = tickSym;                    %ticker symbol
    stock_data(y).Date = date;                        % extract date
    stock_data(y).Open = str2double(open);
     % extract open price
    stock_data(y).High = str2double(h);              % extract high price
    stock_data(y).Low = str2double(l);               % extract low price
    stock_data(y).Close = str2double(close);
     % extract close price

   stock_data(y).Vol = str2double(vol);              % to extract volume
```

```
    stock_data(y).Adj_Close = str2double(a_close);
     % to extract adjustied close data

    y = y + 1;                                  % increment stock index
end

clear date open h l close vol a_close temp status

% update waitbar
waitbar(1/length(tickSym),x)
close(x)                                        % close waitbar
```

To calculate hurst parameters and to extract stock based in their rsquare values.

stock_correlation_H_volume.m:

```
m = 1;
s = [];
sse = [];
slope = [];
slopes = [];                    %To clear all variables from previous run
rsquare = [];
rsquares = [];
fractal_dimension = [];
fractal_dimensions = [];
avg_volume = [];
avg_volumes = [];

for i = 1:length(stocks)

    if(length(stocks(i).Close)<1000)   %Condition to check considerable
number of data points
        continue;
    end
    z = stocks(i).Close;
    y = flipud(z);                  %To reverse the order of prices from
```

```
        if (min(y)==max(y))
            continue;
        end
         max_k = round(0.75*length(y));     %To limit the size of k
        [slope(i),rsquare(i),s(i)] = find_h(y, max_k); %To estimate slope or 'h'
parameter
        avg_volume(i) = mean(abs(stocks(i).Volume));
        fractal_dimension(i) = 2 - slope(i);
        if rsquare(i) < 0.9         %To select stocks that best fit the fractal model
            continue;
        end
        if (isnan(slope(i)))              %To check for NAN values
            continue;
        end
        slopes(m) = slope(i);
        sse(m) = s(i);
        fractal_dimensions(m) = 2-slopes(m);
        rsquares(m) = rsquare(i);
        avg_volumes(m) = mean(abs(stocks(i).Volume));


        m = m+1;
    end
    %To caculate the Pearson Coefficient and p-value
    [pearson_corr, p_value] = corr(fractal_dimensions',avg_volumes')
    plot(fractal_dimensions',avg_volumes','.')  %Scatter plot
```

find_h.m

```
function [m,r,s] = find_h(y,max_k)

assert(max_k>=1); %To assert that max_k is always > 1
assert(max_k<=length(y)); %To assert that max_k is less than sample size
assert(isvector(y)); %To assert that y is existent variable

z = log(y);
correlogram = zeros(1,max_k);
%To find the mean of the absolute difference of the price records with
%time interval k
for k=1:max_k
    correlogram(k) = mean(abs(z((k+1):end)-z(1:(end-k))));
end

x = log(1:max_k);

assert(all(correlogram~=0)); %To assert no NAN values are stored
log_correlogram = log(correlogram);

[m,r,s] = linear_fit(x,log_correlogram); %Invoke function to calculate
'hurst' parameter
```

linear_fit.m

```
  function [a, rsq, sse]=linear_fit(X, Y)
s_x=0; %(s_x is a variable for sum of x-values)
s_y=0; %(s_y is a variable for sum of y-values)
s_xy=0; %(s_xy is a variable for sum of products of x and y-values)
s_x_sq=0; %(s_x_sq is a variable for sum of squares of x-values)
s_y_sq=0; %(s_y_sq is a variable for sum of squares of y-values)
m=size(X, 2); %(m is the number of x and y values)



%Loop to find the sum of residual squares
for i=1:m
    s_x=s_x+X(i);
    s_y=s_y+Y(i);
    s_x_sq=s_x_sq+X(i)^2;
    s_y_sq=s_y_sq+Y(i)^2;
    s_xy=s_xy+X(i)*Y(i);
end

a=(m*s_xy-s_x*s_y)/(m*s_x_sq-s_x^2); %Calculating the slope of the best
fit line
b=(s_x_sq*s_y-s_xy*s_x)/(m*s_x_sq-s_x^2); %Calculating the y-intercept.

yresid = zeros(1,length(Y));
```

```
y = zeros(1,length(Y));

for j=1:m

    y(j) = a*X(j)+b;
    yresid(j) = Y(j)- y(j);

end

sse = 0;
sstot = 0;
rsq = 0;

sse = sum(yresid.^2); %Calculating the sum of squared errors
sstot = (length(Y)-1) * var(Y);
rsq = 1 - sse/sstot;
```